# Cardiac Deformation Lab

Advanced Statistical & Quantitative Analytics Report

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|---|---|---|---|
| **Study Title:** | **Report Test Study e101ad** | **Subject Identifier:** | TEST-REP-01 |
| **Study ID:** | 0fc7d605-68d0-4366-93e9-e2bc69c576c5 | **Session Count:** | **2** analyses |
| **Analysis Scope:** | **Full Image** | **Detected Outliers:** | 0 observations (1.5xIQR) |
| **Report Date:** | 2026-10-18 09:38 UTC | **Computational Engine:** | NumPy / Farnebäck Dense Flow |

## 1. Descriptive Deformation Statistics (Univariate)

| Metric | N | Mean | Median | Std Dev | Min | Max | P25 | P75 | IQR |
|---|---|---|---|---|---|---|---|---|---|
| **Mean Displacement (px)** | 2 | 1.5011 | 1.5011 | 0.0000 | 1.5011 | 1.5011 | 1.5011 | 1.5011 | 0.0000 |
| **Max Displacement (px)** | 2 | 6.6640 | 6.6640 | 0.0000 | 6.6640 | 6.6640 | 6.6640 | 6.6640 | 0.0000 |
| **Anisotropy Index** | 2 | 0.4852 | 0.4852 | 0.0000 | 0.4852 | 0.4852 | 0.4852 | 0.4852 | 0.0000 |
| **Strain Entropy** | 2 | 2.8164 | 2.8164 | 0.0000 | 2.8164 | 2.8164 | 2.8164 | 2.8164 | 0.0000 |
| **Strain Gradient** | 2 | 0.0329 | 0.0329 | 0.0000 | 0.0329 | 0.0329 | 0.0329 | 0.0329 | 0.0000 |
| **Motion Coherence** | 2 | 0.8749 | 0.8749 | 0.0000 | 0.8749 | 0.8749 | 0.8749 | 0.8749 | 0.0000 |
| **Quality Control Score (/100)** | 2 | 98.0000 | 98.0000 | 0.0000 | 98.0000 | 98.0000 | 98.0000 | 98.0000 | 0.0000 |

## 2. Longitudinal Change & Standardized Differences

| Tracked Metric | First | Latest | Delta (Δ) | Rel Change | Std Diff (d) | Trend |
|---|---|---|---|---|---|---|
| **Mean Displacement** | 1.5011 px | 1.5011 px | 0.0000 px | +0.0% | 0.0000 | **Stable** |
| **Max Displacement** | 6.6640 px | 6.6640 px | 0.0000 px | +0.0% | 0.0000 | **Stable** |
| **Anisotropy Index** | 0.4852 | 0.4852 | 0.0000 | +0.0% | 0.0000 | **Stable** |
| **Strain Entropy** | 2.8164 | 2.8164 | 0.0000 | +0.0% | 0.0000 | **Stable** |
| **Strain Gradient** | 0.0329 | 0.0329 | 0.0000 | +0.0% | 0.0000 | **Stable** |
| **Motion Coherence** | 0.8749 | 0.8749 | 0.0000 | +0.0% | 0.0000 | **Stable** |
| **Quality Control Score** | 98.0000 /100 | 98.0000 /100 | 0.0000 /100 | +0.0% | 0.0000 | **Stable** |

## 3. Metric Pearson Correlation Analysis

*Insufficient paired observations (<3 sessions) for Pearson metric correlation.*

## 4. Computational Statistical Outliers (1.5×IQR)

*No computational outliers identified using the 1.5×IQR boundary rule (or insufficient N < 4 observations).*

## 5. Statistical Warnings & Protocol Consistency

• **Note:** Insufficient paired observations (<3 sessions) for Pearson metric correlation matrix.

## 6. Research Methodology & Interpretation Disclaimer

**Computational Analytics Scope:** Descriptive parameters, consecutive change differentials, standardized computational differences (effect sizes), and Pearson correlation matrices reflect algorithmic evaluations of image-space deformation fields under classical continuum kinematics. Statistical outliers reflect mathematical dispersion (1.5×IQR) and do NOT indicate patient health or clinical abnormality.

***RESEARCH PROTOTYPE ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:***

*These statistics summarize computational image-derived measurements and should not be interpreted as clinical findings. This software does NOT provide medical diagnoses, prognosis, treatment monitoring, or clinical risk assessments. This document is intended strictly for mathematical and biomechanical research experimentation.*